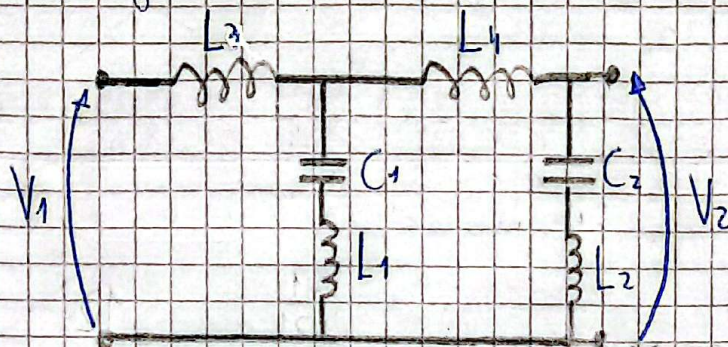


Verificación ejercicio 9



$$C_1 = 2/7 \text{ F} ; C_2 = 24/63 \text{ F} ; L_3 = 3/8 \text{ H}$$

$$L_1 = 7/8 \text{ H} ; L_2 = 7/24 \text{ H} ; L_4 = 9/48 \text{ H}$$

$$\begin{pmatrix} 1 + \frac{SL_3}{SL_1 + 1} & SL_3 \\ \frac{1}{SL_1 + 1} & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 + \frac{SL_4}{SL_2 + 1} & SL_4 \\ \frac{1}{SL_2 + 1} & 1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} = [T]$$

Calculo el parámetro A para obtener luego la transformación de tensión deseada:

$$A = \left(1 + \frac{SL_3}{SL_1 + 1} \right) \left(1 + \frac{SL_4}{SL_2 + 1} \right) + \frac{SL_3}{SL_2 + 1}$$

$$A = \left(\frac{S^2 L_1 C_1 + S^2 L_3 C_1 + 1}{S^2 L_1 C_1 + 1} \right) \left(\frac{S^2 L_2 C_2 + S^2 L_4 C_2 + 1}{S^2 L_2 C_2 + 1} \right) + \frac{S^2 L_3 C_2}{S^2 L_2 C_2 + 1}$$

$$A = \frac{(S^2 L_1 C_1 + S^2 L_3 C_1 + 1)(S^2 L_2 C_2 + S^2 L_4 C_2 + 1) + (S^2 L_3 C_2) \cdot (S^2 L_1 C_1 + 1)}{(S^2 L_1 C_1 + 1)(S^2 L_2 C_2 + 1)}$$

$$A = \frac{(S^4 L_1 C_1 L_2 C_2 + S^4 L_1 C_1 L_4 C_2 + S^4 L_3 C_1 L_2 C_2 + S^4 L_3 C_1 L_4 C_2 + S^4 L_3 C_2 L_1 C_1 + S^2 L_3 C_2 + S^2 L_2 C_2 + S^2 L_4 C_2 + S^2 L_1 C_1 + S^2 L_3 C_1 + 1)}{(S^2 L_1 C_1 + 1)(S^2 L_2 C_2 + 1)}$$

NOTA

$$A = \frac{S^4(L_1C_1L_2C_2 + L_1C_1L_4C_2 + L_3C_1L_2C_2 + L_3C_1L_4C_2 + L_3C_2L_1C_1) + S^2(L_3C_2 + L_2C_2 + L_4C_2 + L_1C_1 + L_3C_1) + 1}{(S^2L_1C_1 + 1)(S^2L_2C_2 + 1)}$$

Reemplazo valores:

$$A = \frac{S^4\left(\frac{7}{8} \cdot \frac{2}{7} \cdot \frac{7}{24} \cdot \frac{24}{63} + \frac{7}{8} \cdot \frac{2}{7} \cdot \frac{91}{48} \cdot \frac{24}{63} + \frac{3}{8} \cdot \frac{2}{7} \cdot \frac{7}{24} \cdot \frac{24}{63} + \frac{3}{8} \cdot \frac{2}{7} \cdot \frac{91}{48} \cdot \frac{24}{63} + \frac{3}{8} \cdot \frac{24}{63} \cdot \frac{7}{8} \cdot \frac{2}{7}\right) + S^2\left(\frac{3}{8} \cdot \frac{24}{63} + \frac{7}{24} \cdot \frac{24}{63} + \frac{91}{48} \cdot \frac{24}{63} + \frac{7}{8} \cdot \frac{2}{7} + \frac{3}{8} \cdot \frac{2}{7}\right) + 1}{(S^2 \frac{7}{8} \cdot \frac{2}{7} + 1)(S^2 \frac{7}{24} \cdot \frac{24}{63} + 1)}$$

$$A = \frac{S^4 \frac{1}{3} + S^2 \frac{4}{3} + 1}{(S^2 \frac{1}{4} + 1)(S^2 \frac{1}{9} + 1)} = \frac{\frac{1}{3}(S^4 + S^2 \cdot 4 + 3)}{\frac{1}{4} \cdot \frac{1}{9}(S^2 + 4)(S^2 + 9)}$$

$$A = 12 \frac{(S^2 + 1)(S^2 + 3)}{(S^2 + 4)(S^2 + 9)}$$

Donde finalmente la transferencia de tensión:

$$A^{-1} = \frac{V_2}{V_1} = \frac{1}{12} \frac{(S^2 + 4)(S^2 + 9)}{(S^2 + 1)(S^2 + 3)}$$